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A market-oriented approach for intermodal network optimisation meeting cost, time and environmental requirements

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ABSTRACT

An increasing number of logistics service providers act as freight integrators in offering one-stop solutions to customers in recent years. Most mathematical models of intermodal network design have been developed with a single objective of cost minimisation. This study focuses on developing innovative approaches in the area of enhanced intermodal network design provided by freight integrators. The overall aim of this paper is to demonstrate an original optimisation model that can be employed by freight integrators to address cost minimisation, transit time minimisation, and carbon footprint to better meet market needs. To achieve the aim, this paper develops a bi-objective optimisation model to minimise cost and transit time for the tactical planning of intermodal container flows with constrained carbon emission. The results and analysis of the example of China offer practical insights on the impact of trade-offs between cost and transit time, and the effect of different carbon emission restrictions on intermodal network design.

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1. Introduction

Containerisation which was invented about sixty years ago has brought logistics and transportation into a new stage. The way of cargo handling standardisation provides a solution which drives different transportation modes towards a seamless process (Thill and Lim, 2010). Hence the practice of intermodal transportation has become prevalent. Container intermodal transport can be well-described as container transport in multimodal chains which link the original nodes of consignors to the destination nodes of consignees so as to offer door-to-door services to customers. At the same time, the rapid development of supply chain management leads to a deeper integration between logistics services. The demarcation between previously exclusive logistics markets is now unclear. One-stop services achieving integration among various stages in supply chains are well welcomed by customers (Lam and Van de Voorde, 2011). Correspondingly, an increasing number of logistics service providers act as freight integrators in offering more integrated solutions to customers. These companies are eager to obtain a bigger piece of cake in a highly competitive environment to achieve the loyalty of their customers (Perez-Labajos and Blanco, 2004). Innovation plays a key role in the dynamic and competitive logistics market to better serve customers (Flint et al., 2005). Therefore, this study focuses on developing innovative approaches in the area of enhanced intermodal network design provided by freight integrators.

The design of intermodal networks has attracted increasing interests in academia. Nevertheless, there remain major literature gaps in this research domain. First, according to a recent literature survey (Lam and Gu, 2013), most mathematical models of intermodal network design have been developed with a single objective of cost minimisation (e.g. Iannone and Thore, 2010; Wang and Yun, 2013). Only few models integrated transit time as another objective (Jula et al. 2006; Zhang et al. 2009; Yang et al. 2011; Zhang et al. 2011), or included multiple objectives (Erera et al. 2005; Wong et al. 2010). Innovative solution methods should be developed to meet the need of a diversified market as some customers prefer lowest freight rates while some others would rather pay more for a faster delivery. Second, many research works focused on container routing and empty container repositioning in sea transportation. However, very few papers considered both sea and land legs together (Min 1991; Kim et al. 2008; Imai et al. 2009; Infante et al. 2009). Third, the issue of environmental protection has become a key concern in sustainable development in the logistics and transportation industries (Lun et al., 2015). Hence there is a growing need to consider carbon footprint in intermodal network design. To the authors' knowledge, we do not find any intermodal network optimisation model testing the effect of different carbon emission levels. Intermodal networks offer a great potential to improve sustainability because railway and inland barge transport generally incur much lower carbon emissions than trucking (Kim and Van Wee, 2014), which is currently a dominant

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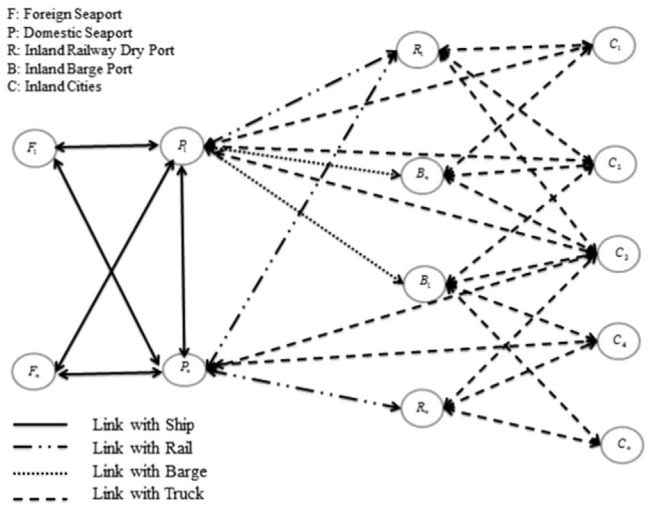


Fig. 1. Port hinterland intermodal container transport network.
Source: Drawn by the authors.

mode in inland transport. This explains why green transport initiatives suggest solutions such as seaports linked with inland dry ports by railway especially double-stack train application, inland barge connections, and using shortest possible initial and final journeys by trucks in intermodal networks (Rahimi et al. 2008; Liao et al. 2009). In general, environmental issues have been increasingly researched, but quantitative models developed to address such significant issues are relatively scarce (Rahimi et al. 2008; Shintani et al. 2010).

In view of the identified research gaps, this paper develops an original optimisation model for the tactical planning of a port hinterland intermodal network. The overall aim of this paper is to demonstrate an enhanced intermodal network design that can be performed by freight integrators to address cost minimisation, transit time minimisation, and carbon footprint to better meet market needs. There are three major inland transportation modes to link shipping, including railway, barge, and truck. These three major transportation modes incur different costs, transit times, and carbon emissions. How to manage the trade-offs among these three aspects is a major challenge to freight and logistics planners, operators and users, because their corresponding management objectives may conflict with each other most of the time (Lam and Lai, 2015; Notteboom, 2010). This study develops an innovative approach to address these trade-offs and to analyse the effect of different carbon emission restrictions on intermodal network design using a case study of China.

The rest of this paper is organised as follows: Section 2 describes the detailed problem and presents model formulation. Section 3 applies the model for a case study regarding China and reports the numerical results. Section 4 discusses managerial implications. Section 5 concludes this research.

2. Problem description and model development

2.1. Port hinterland intermodal network design problem

The model is used for the tactical planning of a port hinterland intermodal network. The detailed network is illustrated in Fig. 1. Containerised freights are shipped from many foreign seaports to many domestic seaports. After containers are discharged at domestic seaports, customs clearance is required before they can be routed through an inland transport network to end-customers' distribution centres at domestic inland cities. Three available inland transportation modes could be chosen, including truck, railway, and barge. Where there are available rail linkages or barge ports nearby, line-haul may be done by rail or barge before last mile delivery by trucking. Without such facilities, containers could also be transported from domestic seaports to end-customers all the way by trucking.

2.2. A bi-objective optimisation model

This model has two objectives: cost minimisation and transit time minimisation. Bi-objective optimisation is more reasonable and practical than single objective optimisation. In real-life situations, decision makers often need to deal with conflicting objectives. Cost and transit time are the two most common considerations in transport planning problems (Lam and Gu, 2013). This model is formulated in a setting that a freight integrator is the decision maker. We assume that the freight integrator uses one type of container ship and there is no capacity constraint for container transport in the sea leg since shipping capacity is not the focus of this study. The main focus is put on the hinterland container transport optimisation and its modal split situation. Also, this assumption fits the current market situation of over-supply in general. Transit times are assumed to be deterministic at all transportation modes. This assumption suits the tactical planning time horizon and the study scope. To analyse the effect of different carbon emission requirements on intermodal network design, carbon emission restrictions set by the government for transport operations are considered as a model constraint. This section presents the model formulation and corresponding explanations are given as follows.

2.2.1. Model formulation

2.2.1.1. Sets.

Set	Description
N	A set of nodes, let $N = F \cup P \cup R \cup B \cup C$, while F stands for foreign seaports, P stands for domestic seaports, R stands for dry ports linked by railway, B stands for barge ports, C stands for inland cities.
A	A set of arcs, let $A = A_{FP} \cup A_{PF} \cup A_{PR} \cup A_{RP} \cup A_{PB} \cup A_{BP} \cup A_{PC} \cup A_{CP} \cup A_{RC} \cup A_{CR} \cup A_{BC} \cup A_{CB}$, For each $(i,j) \in A_{XY}$, (i,j) denotes the arc from $i \in X$ and $j \in Y$, and $X, Y \in \{F, P, R, B, C\}$.

2.2.1.2. Decision variables.

Decision variable	Description
tcm_{ij}	Total container transport quantity from node n_i to n_j in TEUs, $(i,j) \in A$
ecm_{ij}	

	Empty container transport quantity from node n_i to n_j in TEUs, $(i,j) \in A$
lcm_{ij}	Laden container transport quantity from node n_i to n_j in TEUs, $(i,j) \in A$
$vnum_{ij}$	Number of deployed vehicles from node n_i to n_j, $(i,j) \in A$ while $n_i, n_j \notin F$

2.2.1.3. Parameters.

Parameter	Description
G	Average carbon emission limitation of this network per TEU in kg
$CIMCS$	Customs clearance cost for import laden container per TEU
$CEXCS$	Customs clearance cost for export laden container per TEU
$CIMO$	Other cost for import container per TEU, including container inspection quarantine fee, document fee, and port security fee.
$CEXO$	Other cost for export container per TEU, including container inspection quarantine fee, document fee, and port security fee.
EX_{ij}	Container quantity per vehicle on the arc $(i,j) \in A$
FIX_{ij}	Fixed cost per vehicle on the arc $(i,j) \in A$
SPM_i	Container supply quantity of node n_i in TEUs, $n_i \in F \cup C$
DMM_i	Container demand quantity of node n_i in TEUs, $n_i \in F \cup C$
$SPEM_i$	Empty container supply quantity of node n_i in TEUs, $n_i \in F \cup C$
$DMEM_i$	Empty container demand quantity of node n_i in TEUs, $n_i \in F \cup C$
CAP_i	Container throughput capacity of node n_i in TEUs, $n_i \in P \cup R \cup B$
CHC_i	Container handling cost in node n_i per TEU, $n_i \in P \cup R \cup B$
THT_i	Container handling time in node n_i per TEU, $n_i \in P \cup R \cup B$
CST_i	Container storage cost in node n_i per hour per TEU, $n_i \in P \cup R \cup B$
TST_i	Container storage time in node n_i per TEU, $n_i \in P \cup R \cup B$
CTP_{ij}	Transportation cost from node n_i to n_j in US$ per TEU, $(i,j) \in A$
TTP_{ij}	Transportation time from node n_i to n_j in hours per TEU, $(i,j) \in A$
GTP_{ij}	Transportation carbon emission from node n_i to n_j in kg per TEU, $(i,j) \in A$
$AVNUM_{ij}$	Available vehicle number from node n_i to n_j, $(i,j) \in A$ while $n_i, n_j \notin F$

2.2.1.4. Two objective functions:.

$$(1)\ \text{Minimisecost}\,2em = \left(\sum_{(i,j)\in A} CTP_{ij} \times tcm_{ij} + \sum_{(i,j)\in A} 2 \times (CHC_i) \times tcm_{ij} + \sum_{(i,j)\in A, n_i \notin F \cup C} (CST_i \times TST_i) \times tcm_{ij} + \sum_{(i,j)\in A_{FP}} (CIMCS + CIMO) \times lcm_{ij} + \sum_{(i,j)\in A_{PF}} (CEXCS + CEXO) \times lcm_{ij} + \sum_{(i,j)\in A_{FP}} CIMO \times ecm_{ij} + \sum_{(i,j)\in A_{PF}} CEXO \times ecm_{ij} + \sum_{(i,j)\in A, n_i \notin F, n_j \notin F} FIX_{ij} \times vnum_{ij} \right) \div \sum_{i \in F} (SPM_i + DMM_i) \quad (1)$$

$$(2)\ \text{Minimisetime} = \left(\sum_{(i,j)\in A} TTP_{ij} \times tcm_{ij} + \sum_{(i,j)\in A} 2 \times (THT_i) \times tcm_{ij} + \sum_{(i,j)\in A, n_i \notin F \cup C} TST_i \times tcm_{ij} \right) \div \sum_{i \in F} ((SPM_i + DMM_i) \times 24) \quad (2)$$

2.2.1. 5 Constraints:

$$\sum_{(i,j)\in A} (GTP_{ij} \times tcm_{ij}) \div \sum_{i \in F} (SPM_i + DMM_i) \le G \quad (3)$$

$$\sum_{(k,i)\in A} tcm_{ki} = \sum_{(i,j)\in A} tcm_{ij}, \forall n_i \in P \cup R \cup B \quad (4)$$

$$\sum_{(i,j)\in A} tcm_{ij} = SPM_i, \forall n_i \in F \cup C \quad (5)$$

$$\sum_{(i,j)\in A} tcm_{ij} = DMM_j, \forall n_j \in F \cup C \quad (6)$$

$$\sum_{(i,j)\in A_{FP}} ecm_{ij} = SPEM_i, \forall n_i \in F \quad (7)$$

$$\sum_{(i,j)\in A_{PF}} ecm_{ij} = DMEM_j, \forall n_j \in F \quad (8)$$

$$vnum_{ij} \le AVNUM_{ij}, \forall (i,j) \in A, n_i \notin F, n_j \notin F \quad (9)$$

$$tcm_{ij} \le EX_{ij} \times vnum_{ij}, \forall (i,j) \in A, n_i \notin F, n_j \notin F \quad (10)$$

$$\sum_{(i,j)\in A} tcm_{ij} \le CAP_i, \forall n_i \in P \cup R \cup B \quad (11)$$

$$tcm_{ij} = ecm_{ij} + lcm_{ij}, (i,j) \in A_{FP} \cup A_{PF} \quad (12)$$

$$tcm_{ij} \in \mathbf{Z}^+, \forall (i,j) \in A \quad (13)$$

$$ecm_{ij} \in \mathbf{Z}^+, \forall (i,j) \in A_{FP} \cup A_{PF} \quad (14)$$

$$lcm_{ij} \in \mathbf{Z}^+, \forall (i,j) \in A_{FP} \cup A_{PF} \quad (15)$$

$$vnum_{ij} \in \mathbf{Z}^+, \forall (i,j) \in A, n_i, n_j \notin F \quad (16)$$

2.2.2. Corresponding explanations

First of all, in the model, we combine the inbound and outbound container flows in one formula, which means that one formula represents the two-way (inbound and outbound) flows separately. The objective function (1) minimises total unit costs of laden and empty containers flowing through the intermodal network. They include transportation cost, terminal handling cost, storage cost, customs clearance cost, and the fixed cost of using inland vehicles (trucks, trains and barges). The objective function

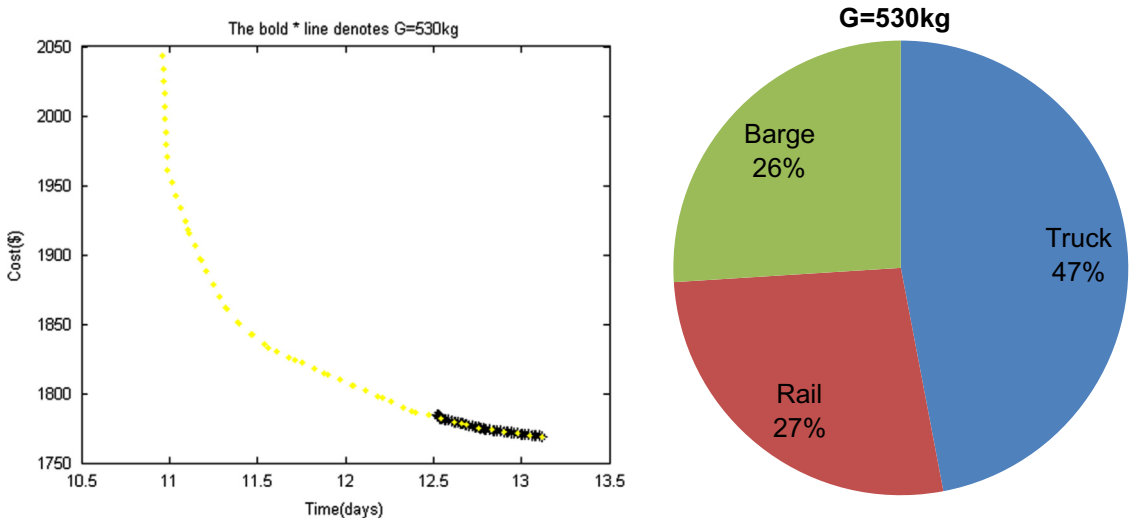


Fig. 2. Modelling results at Carbon Requirement I ($G=530$ kg).
Source: Drawn by the authors from numerical results.

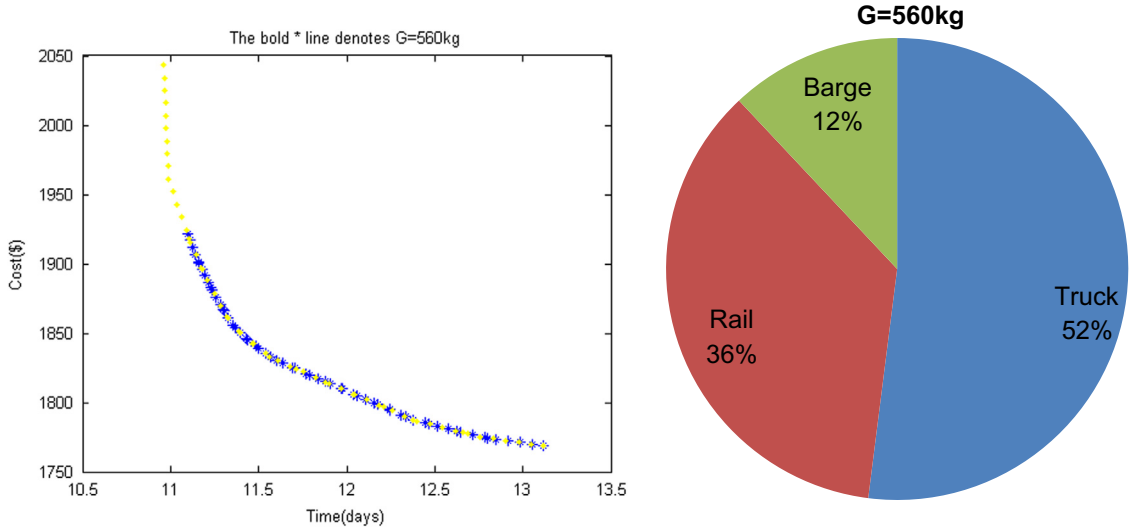


Fig. 3. Modelling results at Carbon Requirement II ($G=560$ kg).
Source: Drawn by the authors from numerical results.

(2) minimises total unit transit times including transportation time, terminal handling time, and storage time. The definition of unit (or average) transit time is used in this model. The total transit time is the sum of transit time of each container routing through the whole network (also including the storage time in each node). Then the total transit time is divided by the total container amount to get the per unit transit time. Constraint (3) represents the carbon footprint emission restrictions set by the government or any regulatory authorities. Constraint (4) balances container in-flows and out-flows at transport nodes. Constraints (5) and (6) are supply and demand constraints of total containers. Constraints (7) and (8) are supply and demand constraints of empty containers. Constraint (9) sets the number of available vehicles in each inland arc. Constraint (10) defines the relationship between container transport quantity and the number of available vehicles in each inland arc. Constraint (11) is a capacity constraint of transport nodes. Constraint (12) defines the relationship of total container transport quantity, laden container transport quantity, and empty container transport quantity in transport arcs. Constraints (13)–(16) define the non-negative characteristics of decision variables.

This model is a bi-objective model which consists of minimising cost and transit time simultaneously. The environmental requirements are handled as a constraint in this model. In order to solve this bi-objective optimisation problem, the single objective optimisation results are needed as the parameters. The two extreme points can be obtained from each single objective optimisation respectively, namely Scenario 1-Cost Minimisation and Scenario 2-Time Minimisation (in below Section 3.2). Pareto Frontiers are obtained by plotting 100 dots which represent different trade-offs between costs and transit times (in below Figs. 2–4). The two extreme points help to give the value range of transit time, then we use fixed transit time dots to get the optimal cost values (Martínez and Eliceche, 2011, Wang et al., 2011).

If some unknown variables are required to be integers, the problem is called a mixed integer linear programming (MILP) problem. These are generally Non-deterministic Polynomial-time hard (NP-hard) (Lodi, 2010). This means the time required to solve even moderate-size problems can easily require a huge computing

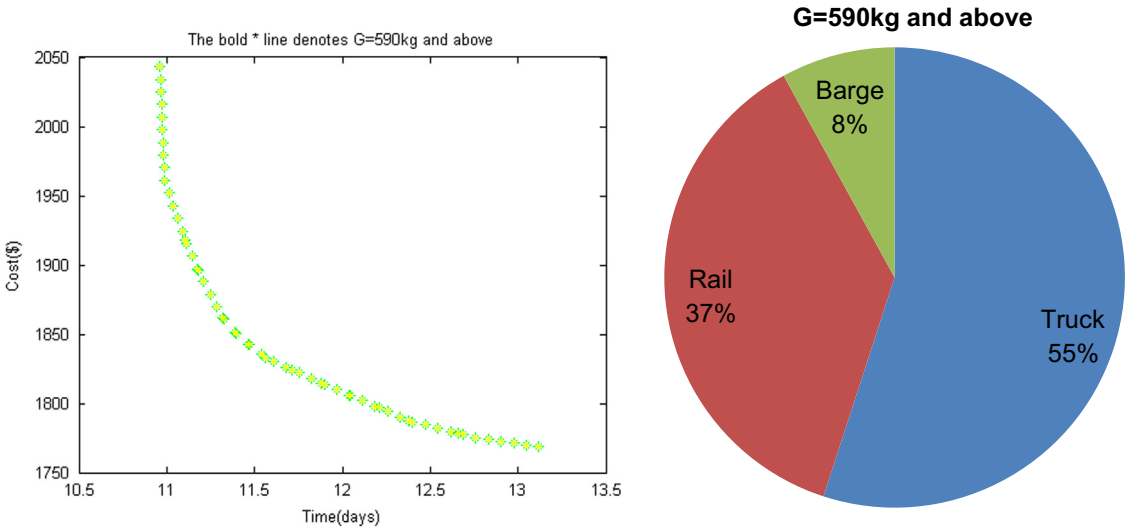


Fig. 4. Modelling results at Carbon Requirement III ($G=590$ kg and above).
Source: Drawn by the authors from numerical results.

Table 1
Main parameters.*Sources:* Compiled from the technical report (TEMS 2008), "BLM-Shipping" professional software, the website of Ministry of Transport of China, and Maersk Line (2013).

	Ship	Rail	Barge	Truck
Variable transportation cost ($/km)	0.19	0.5	0.17	2
Average speed (km/h)	40	70	30	70
Carbon footprint (kg/ton-km)	0.084	0.205	0.084	0.472

Table 2
Mode Usage Rate in three scenarios with different Optimisation Objective portflios.
Source: Numerical results from authors

Mode Usage Rate	Min Cost (Scenario 1)	Min Time (Scenario 2)	Bi-objectives (Min Cost + Min Time) (Scenario 3)
Truck	44%	67%	55%
Barge	33%	0%	8%
Railway	23%	33%	37%

time (Drexl et al., 2009). In this study, the model is a MILP problem. Some meta-heuristic algorithms are needed for solving large scale cases. However, small and medium problem sizes can be solved directly and efficiently through the commercial solver CPLEX 12.4 by a computational time of less than 1 min. Hence we can employ the general MILP solver CPLEX 12.4 to get the Pareto optimal set in this research. The aim of this research is relevant not only to generate the Pareto Frontier, but also to analyse the different compromised solutions of the Pareto Frontier to gain insight that would support an optimal decision making process. Pareto Frontier will be well analysed in the following section. Next, an application of this model is illustrated via a case study in Mainland China.

3. Model application and numerical example

3.1. Description of the example

In this section, we use a numerical example of a freight integrator operating in China to illustrate the bi-objective model and the results. China is a large economy with rapid development and a country with an extensive continent area. China's continued economic rise has caused significant repercussions for the global economy in terms of trade patterns and orientation. On one hand, China's rapid economic growth has brought about the prosperity of its logistics and transportation industries. Its industry development speeds up and promotes its global competitiveness. On the other hand, while China has achieved a remarkable economic performance, there is an increasing need to address environmental reservation for the country's sustainable development (Lam and Notteboom, 2014). In this case, we select the hinterland area of the Yangtze River in Central China as our study's backyard. The Yangtze River is the longest river in China and is one of the main arteries of waterway container traffic. Shanghai Port is located at the mouth of the Yangtze River. Not far away from Shanghai Port, Ningbo–Zhoushan Port is a fast growing port. Shanghai Port and Ningbo–Zhoushan Port are chosen as two domestic seaports in the numerical example. Both ports have achieved extraordinary container traffic growth in the past ten years. Shanghai Port has become the world's busiest container port in terms of throughput volume since 2010. Ningbo–Zhoushan Port was ranked as the 6th container port since 2010 (Marine Department, 2013).

Twenty-five foreign seaports are included in this example. These 25 foreign seaports are heavily involved in Mainland China's international trade. They are Los Angeles, Long Beach, Seattle, Oakland, New York, Boston, Miami, Rotterdam, Antwerp, Hamburg, Le Havre, Kobe, Tokyo, Yokohama, Osaka, Tanjung Pelepas, Port Klang, Singapore, Hong Kong, Busan, Inchon, Gwangyang, Kaohsiung, Keelung, and Taichung. Regarding the hinterland setting, three main inland transport modes are included which are truck, rail, and barge. Since we apply our model to the Yangtze River and its hinterland, barge transport is available under this background. The selected hinterland areas of Shanghai and Ningbo–Zhoushan ports include Shanghai Municipality, Jiangsu Province, Zhejiang Province, Anhui Province, Jiangxi Province, Hubei Province, and Hunan Province. Five big cities are chosen as inland city nodes from each province. Totally, there are 30 selected inland cities. Based on the current situation of infrastructure in Mainland China, we consider five railroad dry ports in Suzhou, Hangzhou, Nanjing, Wuhan and Changsha and three barge ports in Suzhou, Nanjing and Wuhan. Data on distance parameters were collected from various sources, e.g. the distances between seaports were collected mainly from the "BLM-shipping" software, the roadway distances were collected mainly from "www.jdcsww.com", the railway distances were collected mainly from"

Table 3
Transport quantity via inland railroad and barge ports.*Source:* Compiled by the authors from numerical results.

		Carbon Requirement III ($G \geq 590$ kg)	Carbon Requirement I ($G = 530$ kg)
	City	Usage (TEU)	Usage (TEU)
Railway dry port	Suzhou (D0)	2587	4559
	Hangzhou (D1)	1632	3186
	Nanjing (D2)	4110	1676
	Wuhan (D3)	11,339	0
	Changsha (D4)	13,552	15,000
	Subtotal	33,220	24,421
Barge port	Suzhou (B0)	419	1639
	Nanjing (B1)	1394	7126
	Wuhan (B2)	5109	15,000
	Subtotal	6922	23,765
Total by dry ports and barge ports		**40,142**	**48,186**

www.12306.cn", inland waterway distances were collected mainly from "China Ports Year Book".

We assume that there are 80% and 20% empty containers for import and export in Mainland China (Rodrigue 2007). Customs clearance cost per TEU is $120 and $100, respectively for import and export laden containers. Other port handling cost per TEU is $165 and $85, respectively for import and export containers. We suppose one truck carries two TEUs, one rail can carry 100 TEUs, and one barge can carry 500 TEUs per trip. Every selected inland city has a demand of 1500 TEUs and such demand follows a uniform distribution in the planning period. Other main parameters are shown in Table 1.

3.2. Three scenarios with different Optimisation Objective Portfolios

To analyse the trade-offs between total cost and transit time, we generate the results of intermodal network design corresponding to different optimisation objectives for comparison (Table 2). There are three scenarios as follows:

3.2.1. Scenario 1, only total cost minimisation:

If a freight integrator considers only a single objective optimisation ("Min Cost") in intermodal network design, the results are presented in column 1. In this scenario, 44% of containers should be delivered by truck directly from domestic seaports, and 33% and 23% of containers should be transported by barge and railway, respectively.

3.2.2. Scenario 2, only transit time minimisation:

If a freight integrator only conducts "Min Time" optimisation, results are shown in column 2. In this scenario, 67% of containers should be delivered by truck directly from domestic seaports, and 33% of containers should be transported by railway. Barge is not used due to longest transit time.

3.2.3. Scenario 3, bi-objective optimisation:

Column 3 shows the bi-objective optimisation results. In such a scenario, 55% of containers should be delivered by truck directly from domestic seaports, and 8% and 37% of containers should be transported by barge and railway, respectively.

When cost minimisation is the only objective, the barge transport mode which generates the lowest carbon emission is used for 33% of containers, representing a much higher percentage than the case of the other two scenarios. The results show that an economic objective and environmental protection in intermodal transport can be complementary to each other. When transit time minimisation is the only objective, trucking which generates the highest carbon emission is the most preferred transportation mode. Therefore, transit time, not cost, is a major reason for using transport modes that are less environmentally friendly. When we consider these two objectives together, there is a trade-off for the transport mode choice.

Through careful examination, it can be noticed that in all the scenarios, the usage rate of truck is still the highest while the usage rate of barge is relatively low. It is because in this specific case study of China's Yangtze River basin, many inland cities do not have barge links or railway links to seaports. According to the data, about 67% inland cities have road links (truck) to seaports, while the rate of inland waterway links (barge) is 20% and the rate of railway links is 28%. This represents an authentication of the model formulation since our optimisation results of modal split are able to reflect the existing infrastructure situation in this example. The results also reveal that intermodal network design in the Yangtze River basin is restricted by its infrastructure which is much road transport dependent. There is room for enhancing intermodal transport infrastructure for sustainable development. More discussion will be given in Section 4.

In the sea leg, optimisation objectives also affect cargo routing. Shanghai Port is an established international shipping hub. For import routes to the region, many shipping lines call Shanghai Port first and then Ningbo–Zhoushan Port, or link Shanghai and Ningbo–Zhoushan ports with feeder services. The transit time for a ship/barge between Shanghai and Ningbo–Zhoushan ports is about one day. If a customer in Ningbo wants lowest transportation cost, his containers should be discharged at Ningbo–Zhoushan Port. However, if he needs to receive his cargoes fast, containers must be discharged at Shanghai Port and then trucked to Ningbo within several hours.

3.3. Sensitivity analysis in carbon emission restrictions

According to Section 2.2, G represents the average carbon emission limitation per TEU which reflects policy makers' restriction on carbon emission. From the results generated by the optimisation model in this numerical example, when G is less than 530 kg, there is no feasible solution. In order to illustrate the impact of G value, we obtain modelling results at three sets of G values when a feasible solution exists. In Carbon Requirements I and II, G values are set as 530 kg and 560 kg, respectively. In Carbon Requirement III, G value is set as 590 kg or above. Scenario III means that when G value is greater than 590 kg, the change of G value will not impact the range of the Pareto Frontier.

Figs. 2–4 present the numerical results at these three Carbon Requirements. In each figure, Pareto Frontier is on the left side and container distribution chart by transportation modes is on the right side. Pareto Frontiers are obtained by plotting 100 dots which

represent different trade-offs between costs and transit times. We use fixed time dots to optimise cost objective. For example, following a uniform distribution, 100 transit times are selected within the feasible solution space, and they are then set as a model constraint to obtain cost optimal results. After obtaining the average of the optimal results, the modal split percentage is shown on the containers distribution chart by transportation modes in Figs. 2–4.

Based on the analysis, it can be noted that the usage of barge increases when *G* value decreases. The usage of truck increases when the *G* value increases. This finding is in line with the common understanding that barge is the most environmentally friendly and truck is the least environmentally friendly mode. If customers are more concerned about environmental sustainability, a freight integrator can employ this enhanced design approach for analysis and to reduce the *G* value to the required level. The usage of inland railways and waterways may be increased where the geographical conditions allow. This will help reduce last-mile trucking mileage in order to build green intermodal networks.

Importantly, we also find that a small change in the *G* value will have a significant impact on the Pareto Frontier range. The reason is that majority of carbon emission is from the sea transport. Its transport distance contains a major share of the total intermodal chain's mileage. Therefore, due to the sensitivity of the change in *G*, it is crucial to address environmental concerns in intermodal network design. Reducing the trucking mileage for inland transport is not only good for environment protection, but also saves cost because it is cheaper to transport cargoes over long distance by railway or waterway. To realise the benefits, however, there must be substantial improvements in inland intermodal transport infrastructures, including not only inland railways and waterways (as links), but also dry ports and barge ports (as nodes).

For more detailed intermodal network design, results on transport quantity via inland railroad and barge ports in Carbon Requirements I and III are generated (Table 3). These two settings are the two extreme situations in this numerical example. Carbon Requirement I has a *G* value of 530 kg, which represents a stringent carbon emission limit. Carbon Requirement III has a *G* value greater than or equal to 590 kg, which represents a very loose carbon footprint restriction that does not impact transportation mode choices.

In Carbon Requirement I, there are more containers handled by inland railroad and barge ports on the whole because of a tighter carbon footprint restriction. However, total containers transported by railroad decrease as barge becomes dominant for long distance transport. For example, no more containers are transported via Wuhan railroad port (D3) in Carbon Requirement I. However, containers transported via Wuhan barge port (B2) surge dramatically to 15,000 TEUs.

In Carbon Requirement III, Wuhan (D3), Changsha (D4) and Wuhan (B2) are the nodes with the longest transport distances to the two domestic seaports of Shanghai and Ningbo–Zhoushan. They handle a much larger amount of containers than transport nodes which are nearer to domestic seaports. This result means that there is a higher demand for the use of railroad dry ports and barge ports in locations which are far from coastal seaports, even if there are loose restrictions on greenhouse gas emissions.

4. Implications for practice

Practical implications for a more market-oriented intermodal network design can be drawn from the research findings. A growing number of transport and logistics operators as freight integrators now provide one-stop services to their land-locked end-customers. Different freight integrators have different business operation modes. Some freight integrators operate their own inland transport network, while some others prefer outsourcing part of their transport operations. In both cases, freight integrators need to understand how to optimise transport planning based on customer requirements and infrastructure settings (Lam and Dai, 2015). As mentioned in the above sections, under the market driven business environment, they have to optimise total cost and also meet transit time requirements given by customers. At the same time, freight integrators are required to comply with carbon footprint restrictions. As discussed in the above sensitivity analysis, the carbon emission limit in the intermodal network can be adjusted to suit different situations. The original model developed by this study can provide freight integrators with an enhanced and more flexible approach to meet diversified customer needs and environmental regulatory requirements together. This will enable freight integrators to gain a competitive edge in a dynamic market.

Furthermore, in intermodal network design, freight integrators can use the optimisation model for investment and service planning before they enter a foreign market. In the numerical example, China's Yangtze River serves as a natural corridor linking Western China and Eastern China. There is a major benefit of reducing the pollution in the transport sector in Mainland China if barging is further developed and utilised. The Chinese government has been promoting the development of Western and Central China by offering various investment incentives. In fact, the Chinese government has realised that insufficient transport infrastructure hinders the economic development of the western and central regions. Yu et al. (2012) discover that since 1994 the transport infrastructure growth of Mainland China has surpassed its GDP growth and such a gap is bigger from then on. Barge intermodal transport along the Yangtze River from the upstream Chongqing to downstream Shanghai will deserve more attention. More satellite barge terminals will be established along the Yangtze River, while some maintenance works will be improved, such as dredging, widening and deepening the waterway channels accordingly. In comparison with railways and highways, the development of the Yangtze River waterways would require less time and capital investment. Such a development will also generate substantial economic benefits for export oriented manufacturers in the hinterland due to its cost effectiveness in serving trade. As long as inland waterways are available, barge is the most preferred transportation mode to lower cost and carbon emissions.

Where natural waterways are not available, governments need to consider the development of railway links and dry ports as a priority. The railway system in Mainland China is owned by the state and there is little involvement of private investment. Although many years ago some scholars recommended introducing private investment in Chinese Railways in order to strengthen competition (Rong and Bouf, 2005), the Chinese Railway still maintains its own operation pattern today. A large portion of Chinese railway capacity is used for passenger transport due to the large population size, thus limited capacity can be deployed for its freight transport. Nevertheless, the Chinese government has recognised this problem, and has put much effort into its “High Speed Rail” system which can help release more capacity into its freight transport (Guo 2010). Further reforms of the railway system in Mainland China are on the way. Although this development of the railway system is controlled by the Chinese central government, it presents an opportunity for dry port development in Mainland China. Many local governments have expressed a desire to build dry ports themselves by attracting private investment, which means that freight integrators can invest into inland infrastructures. This will give a lot of opportunities to freight integrators to consolidate and strengthen their intermodal networks. In order to avoid over-development, local governments in

Mainland China and relevant private investors should adhere to good-practice planning for the development of dry ports.

Trucking has the shortest transit time but incurs the highest cost and carbon emissions. Unless transit time is a major concern from customers, trucking should not be encouraged for long-distance transport. Barge or railway intermodal transport provides opportunities to reduce the ratio of last-mile trucking mileage to the total transport mileage. This not only brings about economic benefits, but also better preserves the environment. Tighter environmental regulations from governments on carbon emissions favour barge and rail transport, and freight integrators can also obtain economic benefits and good reputations with proper intermodal network planning.

Based on these new developments, freight integrators can study the feasibility of an optimal intermodal network design using the model and conduct planning in advance. More environmentally friendly transport modes require the development of supporting infrastructure including inland waterways, barge ports, railways and dry ports. This requires the joint efforts of governments and freight integrators.

5. Conclusions and future research suggestions

This research has provided an enhanced approach to the intermodal network design problem from the tactical level based on an optimisation model. Freight integrators have ventured into offering highly complicated solutions to improve their competitiveness. This has resulted in an increasing need of simultaneous optimisation of cargo flows in both sea and land legs. The logistics market is much diversified. Some shippers require lowest cost while some others need a fast delivery. Freight integrators also have to address carbon footprint requirements in view of rising environmental concerns. To handle these challenges, this research proposes a bi-objective optimisation model to minimise cost and transit time for port hinterland intermodal network optimisation. The environmental aspect is operationalised by means of including the carbon footprint requirement as a constraint in this model. The study also quantifies the impact of changing carbon emission restrictions on intermodal network design.

The paper has presented a numerical example of import and export container flows to and from inland China and provides four major findings. First, freight integrators should utilise barging as the primary transportation mode to lower cost and carbon emissions whenever inland waterways are available. Second, freight integrators are recommended to use trucking selectively especially for long distance routes since this transportation mode incurs highest cost and carbon emissions. Nonetheless, trucking is an appropriate mode when transit time is a major concern from customers. Third, barge and railway intermodal transport can lower the ratio of last-mile trucking mileage to the total transport mileage. This is an economical solution whilst benefiting the environment. Fourth, tighter environmental regulations on carbon emissions favour barge and rail transport, and a small change in the carbon emission limit will have a significant impact on the range of feasible solutions. Therefore, freight integrators should adopt a more market-oriented approach by flexibly including cost, time and carbon footprint requirements in their intermodal network planning.

There are several contributions made by this paper not only to the academic circle but also the industry. For a scientific contribution, this model is original and fills the gaps in the literature. The model optimises port hinterland intermodal container flows with the consideration of both cost and transit time. Importantly, this study is different from the literature in a way that it quantifies the impact of carbon footprint requirements on intermodal network planning. Also, model application suggests practical insights and implications. The numerical example provides an illustration of the intermodal network design in China. Furthermore, the results can provide a useful reference for other countries with a large continent area. The main contribution of this paper is that this model is beneficial not only to freight operators and one-stop logistics service providers for their innovative service design, but also offers insights for government policy makers to develop inland transport infrastructures in view of cost, time, and environmental considerations.

Same as other studies, this paper contains some limitations and future extensions could be done. First, future research may include sailing speed and bunker consumption in the sea leg as variables. Second, transit times can be modelled to incorporate stochastic features for considering uncertainties. Third, for the intermodal transport planning at an operational level, we are extending the model to consider the routing of individual container origin-destination pairs to meet their cost and transit time requirements. In fact, there are other factors that may affect global intermodal network planning, such as transshipment time of each intermediate node, size and operation efficiency of each node. In our tactical planning model, we have simplified some factors which would not affect the tactical results for managerial implications. In future, the extended model for an operational level optimisation should consider as many other factors as possible.

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